

1. The external force is given by

$$F(t) = m\alpha [\Theta(t) - \Theta(t - T')] \quad (1)$$

To obtain the acceleration, we use the integral equation from problem 16.10

$$\dot{v}(t) = \frac{1}{m} \int_0^\infty e^{-s} F(t + \tau s) ds \quad (2)$$

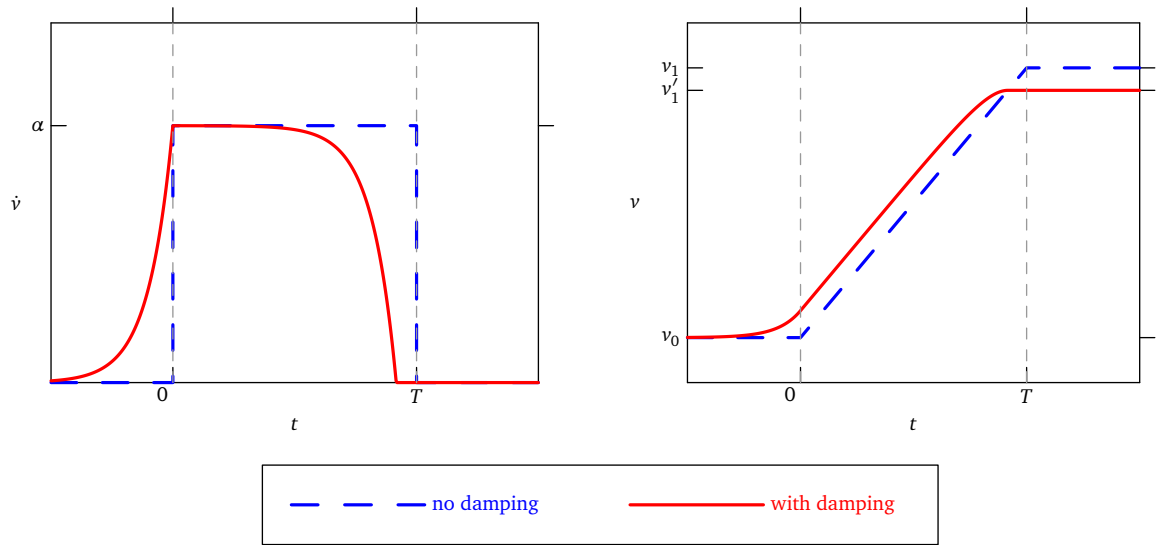
The range of the integral is $[0, \infty) \cap [-t/\tau, (T' - t)/\tau]$, of which the integral is

$$\dot{v}(t) = \alpha \times \begin{cases} e^{t/\tau} - e^{(t-T')/\tau} & t < 0 \\ 1 - e^{-(T'-t)/\tau} & 0 \leq t \leq T' \\ 0 & t > T' \end{cases} \quad (3)$$

Integrating (3) from the incoming state $v(-\infty) = v_0$ gives the velocity

$$v(t) = \begin{cases} v_0 + \alpha\tau(1 - e^{-T'/\tau})e^{t/\tau} & t < 0 \\ v_0 + \alpha\tau + \alpha t - \alpha\tau e^{-(T'-t)/\tau} & 0 \leq t \leq T' \\ v_0 + \alpha T' & t > T' \end{cases} \quad (4)$$

The particle pre-accelerates over a window $\sim \tau$ before entering the gap and the acceleration tapers to zero over $\sim \tau$ before exit. so the emergent velocity is $v'_1 = v(T') = v_0 + \alpha T'$. Because energy is radiated, $v'_1 < v_1$.



2. To lowest order in τ it is cleaner to use the truncated form (16.10)

$$m\dot{v} = F + \tau\ddot{F} \quad (5)$$

Inside the gap F is constant so $\dot{v} = \alpha$, exactly as without damping. The damping acts only at the edges, where

$$\ddot{F} = m\alpha [\delta(t) - \delta(t - T')] \quad (6)$$

Integrating across entry and exit,

$$v(0^+) = v_0 + \alpha\tau \quad v(T'^+) = v(T'^-) - \alpha\tau \quad (7)$$

consistent with the smooth profile (4). Writing the relation between T' and d , we have

$$d = v(0^+)T' + \frac{\alpha T'^2}{2} \quad (8)$$

of which the solution is

$$\begin{aligned}
T' &= -\frac{v(0^+)}{\alpha} + \sqrt{\frac{v(0^+)^2}{\alpha^2} + \frac{2d}{\alpha}} \\
&= \frac{-(v_0 + \alpha\tau)}{\alpha} + \frac{1}{\alpha} \sqrt{(v_0 + \alpha\tau)^2 + 2\alpha d} & v_1 = \sqrt{v_0^2 + 2\alpha d} \\
&= -\frac{v_0}{\alpha} - \tau + \frac{v_1}{\alpha} \sqrt{1 + \frac{2\alpha\tau v_0 + \alpha^2\tau^2}{v_1^2}} \\
&\approx \left(\frac{v_1 - v_0}{\alpha}\right) - \tau + \frac{v_0}{v_1} \tau \\
&= T - \tau \left(1 - \frac{v_0}{v_1}\right)
\end{aligned} \tag{9}$$

By (7), the emergent velocity is

$$v'_1 = v(T'^+) = v(0^+) + \alpha T' - \alpha\tau = v_0 + \alpha T' = v_1 - \frac{\alpha^2\tau}{v_1} T \tag{10}$$

3. The work done by the applied field is

$$W = mad = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_0^2 \tag{11}$$

The energy radiated is the integrated Larmor power,

$$E_{\text{rad}} = \frac{2e^2}{3c^3} \int \dot{v}^2 dt = m\tau \int \dot{v}^2 dt \approx m\tau\alpha^2 T \tag{12}$$

where we have ignored the contribution from the precursor, which is of order $O(\tau^2)$ since by (3)

$$\int_{-\infty}^0 \dot{v}^2 dt \approx \alpha^2 \int_{-\infty}^0 e^{2t/\tau} dt = \frac{\alpha^2\tau}{2} \tag{13}$$

Using (10), the change in kinetic energy is

$$\Delta K = \frac{1}{2}mv_1'^2 - \frac{1}{2}mv_0^2 \approx \frac{1}{2}mv_1^2 - \frac{1}{2}mv_0^2 - m\alpha^2\tau T = mad - m\alpha^2\tau T = W - E_{\text{rad}} \tag{14}$$